

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

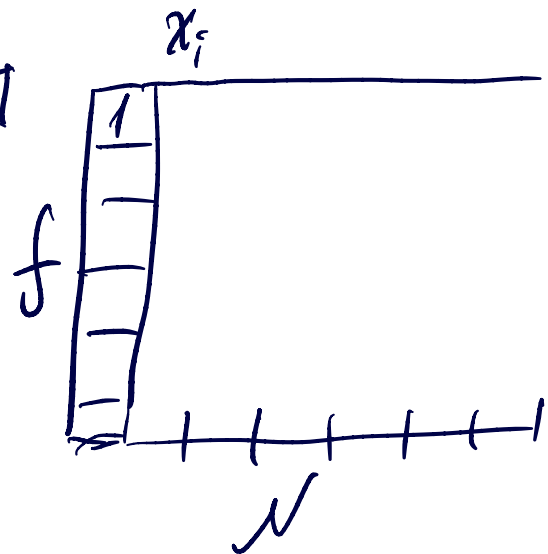
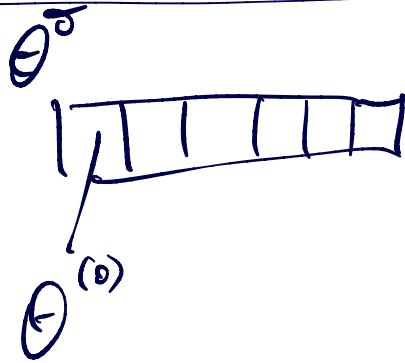
$$P(A, B) = P(A|B) P(B) = P(B|A) P(A)$$

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

$$y = kx + 6$$

$$y = \theta^T x$$

$$\hat{y} = k_1 x_1 + k_2 x_2 + k_3 x_3 + k_0 x_0$$



$$X$$

	f_1	f_2	f_3
$i \downarrow$			
N			

$$X = X^T$$

$$\hat{y} = \Theta^T X$$

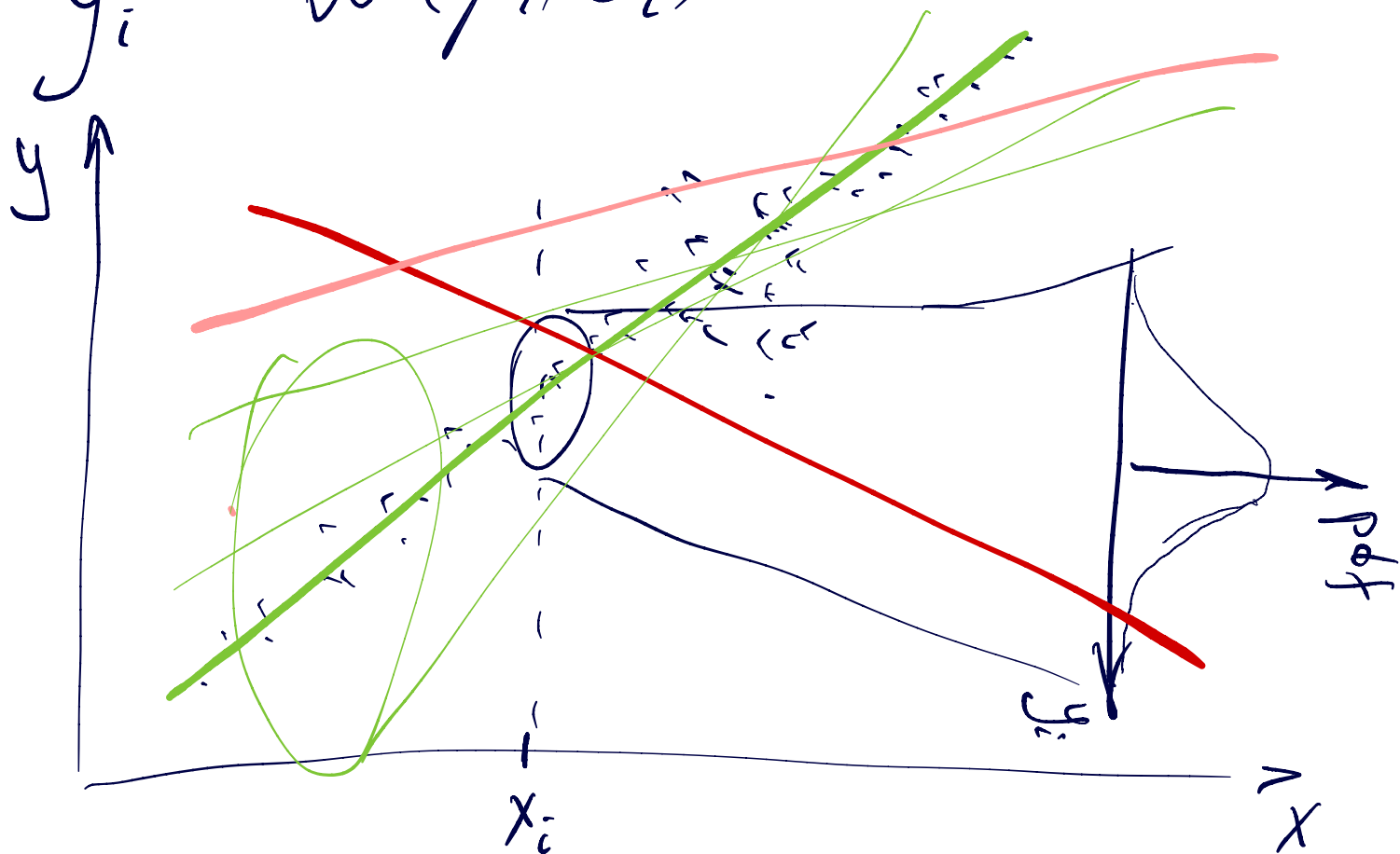
$$f_y \times N = 1 \times f_x \cdot f_x \times N$$

$$f_y \times N = 1 \times N$$

$$\begin{aligned} \Theta &: f_y \times f_x \\ y &: f_y \times N \\ x &: f_x \times N \end{aligned}$$

LINE

$$y_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$$



$$\textcircled{1} L: \mu_i = \theta^T x_i$$

$$\textcircled{2} I: \text{i.i.d.}$$

$$\textcircled{3} N: y_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$$

$$\textcircled{4} E: \sigma_i = \sigma$$

$$P(\theta | x, y) = \frac{P(x, y | \theta) P(\theta)}{P(x, y)}$$

$$P(x_i, y_i | \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(\mu_i - y_i)^2}{2\sigma^2}}$$

$$P(x, y | \theta) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(\mu_i - y_i)^2}{2\sigma^2}}$$

$$\theta^* = \underset{\theta}{\operatorname{argmax}} P(x, y | \theta)$$

$$\ell(\theta | \theta) = \ln \prod \frac{1}{\sqrt{\dots}} e^{-\dots} =$$

$$= \sum \left(\ln \frac{1}{\sqrt{\dots}} e^{-\dots} \right) =$$

$$= \sum_i \left[\ln \frac{1}{\sqrt{\dots}} + \ln e^{-\dots} \right] =$$

$$= N \frac{1}{\sqrt{2\pi\sigma^2}} + \sum_{i=1}^N \frac{(\mu_i - y_i)^2}{2\sigma^2}$$

$$\theta^* = \underset{\theta}{\operatorname{argmax}} \ell(\mathcal{T}|\theta) =$$

$$= \underset{\theta}{\operatorname{argmax}} \left[\cancel{N \frac{1}{\sqrt{2\pi\sigma^2}}} - \frac{1}{2\sigma^2} \sum_{i=1}^N (\mu_i - y_i)^2 \right]$$

$$= \underset{\theta}{\operatorname{argmin}} \sum_{i=1}^N (\mu_i - y_i)^2$$

$$\mathcal{L}(\mathcal{T}, \theta) = (\theta^T X - Y)^T (\theta^T X - Y) =$$

$$= Y^T Y - 2(\theta^T X)^T Y + (\theta^T X)^T \theta^T X$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = -2 X^T Y + 2 \theta^T X^T X$$

$$-2X^T Y + 2\Theta^T X^T X = 0$$

$$\Theta^T X^T X = X^T Y$$

$$\Theta^* = (X^T Y (X^T X)^{-1})^T$$

$$P(\theta|Y) = \frac{P(Y|\theta) P(\theta)}{P(Y)}$$

$P(\theta|Y)$ - апостериорная в-сть θ

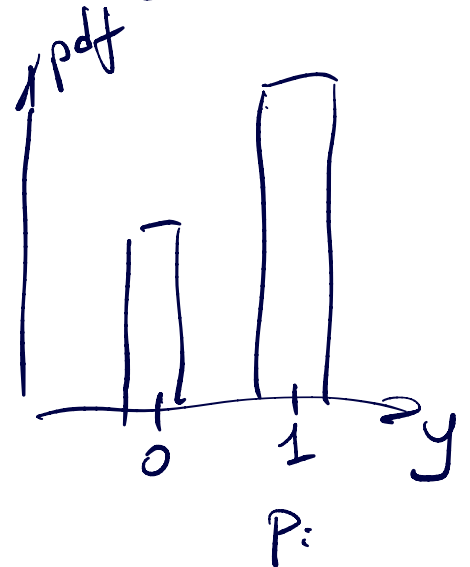
$P(Y|\theta)$ - правдоподобие выборки Y

$P(\theta)$ - априорное рас-е θ

$P(Y)$ - (evidence), вероятность данных

$$x_i : y_i \sim B(p_i)$$

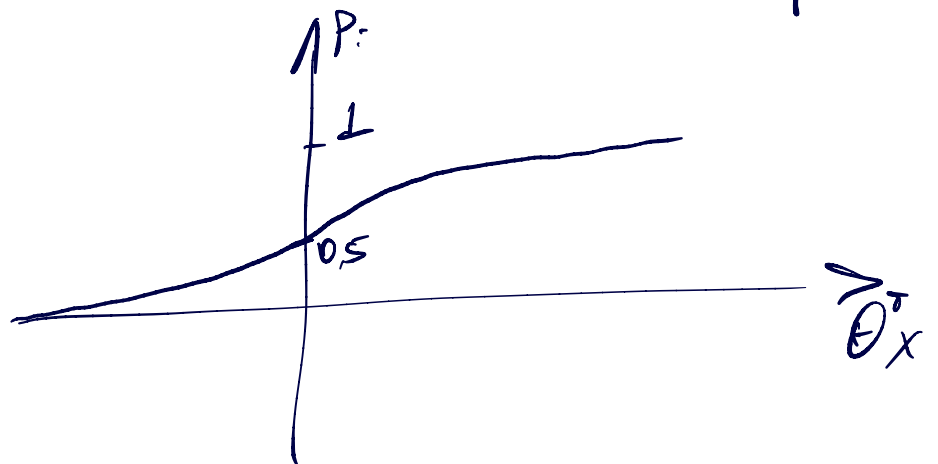
$$P(x_i, y_i | \theta) = p_i$$



$$p_i = \phi(\theta^T x)$$

$$\phi = G$$

$$G(z) = \frac{1}{1 + e^{-z}}$$



$$P(x_i, y_i | \theta) = p_i^{y_i} \cdot (1 - p_i)^{1 - y_i}$$

$$P(\mathcal{T} | \theta) = \prod_{i=1}^N P(x_i, y_i | \theta)$$

$$\begin{aligned} \ell(\mathcal{T} | \theta) &= \sum_{i=1}^N \ln(p_i^{y_i} \cdot (1 - p_i)^{1 - y_i}) = \\ &= \sum_{i=1}^N y_i \ln p_i + (1 - y_i) \ln(1 - p_i) \end{aligned}$$

$$\mathcal{L}(\mathcal{T} | \theta) = - \sum_{i=1}^N y_i \ln p_i + (1 - y_i) \ln(1 - p_i)$$

$$\theta^* = \arg \min_{\theta} \mathcal{L}(\mathcal{T} | \theta)$$

$$p_i = \sigma(\theta^T x_i) = \frac{1}{1 + e^{-\theta^T x_i}}$$